

AI Personal Style & Wellness Advisor

Complete Full Stack + ML + Requirement + Architecture Master Document

This document contains the complete project planning and development structure for the AI Personal Style & Wellness Advisor system.

DOCUMENT STRUCTURE

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PROJECT REQUIREMENT DOCUMENT

PROJECT: AI PERSONAL STYLE & WELLNESS ADVISOR

1. TEAM MEMBERS

Name	Role	Responsibility	Tools Used
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Your Name	Team Leader	Full Stack Development, Project Management	VS Code, GitHub
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Team Member 2	Frontend Developer	React UI, Responsive Design	React.js, Tailwind CSS
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Team Member 3	Backend Developer	API Development, Database	FastAPI, SQLite
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Team Member 4	ML Engineer	Face Detection & Recommendation Models	OpenCV, MediaPipe, TensorFlow
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2. PROJECT IDEA

What is your project?

AI Personal Style & Wellness Advisor is an AI-powered web application that analyzes a user's face shape, facial features, style preferences, and profession/interests to provide personalized fashion and wellness recommendations.

The system uses Computer Vision and Machine Learning to:

- Detect face shape
- Analyze facial proportions
- Recommend hairstyles
- Suggest outfits
- Recommend glasses/accessories

- Provide color palette suggestions
- Give wellness and grooming tips

The platform helps users improve personal styling and confidence using AI-based recommendations.

What problem does it solve?

Many people struggle with:

- Choosing suitable hairstyles
- Selecting outfits that match their appearance
- Finding proper fashion combinations
- Understanding which accessories suit their face shape
- Identifying personal style preferences

Most users rely on trial and error or generic online suggestions.

The AI Personal Style & Wellness Advisor solves this by providing:

- Personalized styling recommendations
- AI-based face analysis
- Smart outfit suggestions
- Fashion recommendations based on appearance and preferences

This saves time and improves confidence and personalization.

Who will use it?

Public Users

- Upload photos
- Receive AI analysis
- Get style recommendations

- Save recommendation history

Admin

- Manage styles and recommendations
- Monitor users and analytics
- Update recommendation datasets

3. FEATURES LIST

Must Have (Core Features)

User Registration & Login

- Secure account creation
- JWT authentication
- Profile management

Face Detection System

- Detect face from uploaded image
- Facial landmark extraction
- Face alignment

Face Shape Classification

The system predicts:

- Oval
- Round

- Square
- Diamond
- Heart
- Rectangle

Hairstyle Recommendation System

AI recommends:

- Haircuts
- Hair volume styles
- Beard styles

based on:

- Face shape
- Facial structure

Outfit Recommendation System

The system suggests:

- Casual outfits
- Formal outfits
- Streetwear
- Smart casual styles
- Athletic wear

based on:

- User preference
- Profession
- Occasion

- Appearance

Glasses & Accessories Recommendation

Suggests:

- Frame shapes
- Accessory styles
- Fashion compatibility

Color Palette Analysis

Analyzes:

- Skin tone
- Contrast level
- Hair color

Suggests:

- Suitable clothing colors
- Seasonal palettes

Recommendation History

Stores:

- Previous analyses
- Previous recommendations
- Saved styles

Responsive Dashboard

- Mobile-friendly UI
- Personalized dashboard
- Recommendation overview

Admin Panel

Admin can:

- Add style categories
- Manage recommendations
- View analytics
- Monitor user activities

Nice to Have (Extra Features)

- Virtual hairstyle preview
- AI fashion chatbot
- Real-time webcam analysis
- Outfit try-on simulation
- Weather-based outfit suggestions
- AI wardrobe organizer
- Social sharing

4. TECH STACK

Layer	Technology	Purpose
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Frontend	React.js	User interface
Styling	Tailwind CSS	Responsive UI
Backend	FastAPI	REST API development
Database	SQLite / MongoDB	Store user & recommendation data
ML Framework	TensorFlow / PyTorch	ML model training
Computer Vision	OpenCV, MediaPipe	Face analysis
API Documentation	Swagger UI	API testing/docs
Version Control	GitHub	Team collaboration
Backend Deployment	Render	Hosting backend
Frontend Deployment	Vercel	Hosting frontend

5. PAGES / SCREENS

User Pages

1. Home Page

- Project introduction
- Feature overview
- Upload image button

2. Register Page

- User sign-up form
- Name, email, password

3. Login Page

- JWT authentication
- Secure login

4. User Dashboard

- Welcome section
- Quick recommendation access
- Previous history

5. Upload Photo Page

- Upload image
- Webcam capture

6. Face Analysis Result Page

Displays:

- Face shape
- Facial proportions
- Symmetry details

7. Hairstyle Recommendation Page

Displays:

- Suggested hairstyles
- Beard recommendations

8. Outfit Recommendation Page

Displays:

- Outfit suggestions
- Fashion combinations
- Style categories

9. Color Palette Page

Displays:

- Recommended colors
- Matching palettes

10. Recommendation History Page

Shows:

- Previous analyses
- Saved recommendations

11. Profile Page

- Edit user information
- Update preferences

12. About Page

- Project details
- AI workflow explanation

13. Contact / Support Page

- User support form
- FAQs

Admin Pages

14. Admin Login Page

15. Admin Dashboard

Displays:

- Total users
- Recommendation analytics
- Usage statistics

16. Manage Users Page

Admin can:

- View users
- Remove users
- Manage profiles

17. Manage Recommendations Page

Admin can:

- Add new style rules
- Update recommendations

18. AI Monitoring Page

Monitor:

- Prediction performance
- Recommendation accuracy

19. Feedback Management Page

View:

- User reviews
- Ratings
- Complaints

20. Settings Page

- Theme settings
- Notification settings
- Language preferences

6. DATABASE TABLES

1. users

Column	Data Type	Description
-----	-----	-----
user_id	INT (Primary Key)	Unique user ID
name	VARCHAR(100)	User full name
email	VARCHAR(150)	User email
password	VARCHAR(255)	Encrypted password
role	ENUM('user','admin')	User role
created_at	DATETIME	Account creation time

2. face_analysis

Column	Data Type	Description
-----	-----	-----
analysis_id	INT (Primary Key)	Unique analysis ID
user_id	INT (Foreign Key)	Related user
face_shape	VARCHAR(50)	Predicted face shape
jawline_score	FLOAT	Jawline analysis
symmetry_score	FLOAT	Face symmetry
skin_tone	VARCHAR(50)	Skin tone category
created_at	DATETIME	Analysis timestamp

3. recommendations

Column	Data Type	Description
-----	-----	-----
recommendation_id	INT (Primary Key)	Recommendation ID
analysis_id	INT (Foreign Key)	Related analysis
hairstyle	TEXT	Suggested hairstyle

outfit_style	TEXT	Suggested outfit
glasses_style	TEXT	Suggested glasses
color_palette	TEXT	Suggested colors
created_at	DATETIME	Recommendation time

4. user_preferences

Column	Data Type	Description
-----	-----	-----
preference_id	INT (Primary Key)	Preference ID
user_id	INT (Foreign Key)	Related user
favorite_style	VARCHAR(100)	Preferred style
profession	VARCHAR(100)	User profession
occasion_preference	VARCHAR(100)	Preferred occasion

5. feedback

Column	Data Type	Description
-----	-----	-----
feedback_id	INT (Primary Key)	Feedback ID
user_id	INT (Foreign Key)	Related user
rating	INT	User rating
comments	TEXT	Feedback message
created_at	DATETIME	Submission time

7. API DESIGN

Method	Endpoint	Description	Who Can Use
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	-----		-----		-----		-----	
	POST		/api/auth/register		Register user		Anyone	
	POST		/api/auth/login		Login user		Anyone	
	GET		/api/users/profile		Get user profile		Logged-in User	
	PUT		/api/users/profile		Update profile		Logged-in User	
	POST		/api/analyze-face		Analyze face image		Logged-in User	
	GET		/api/recommend/hairstyle		Get hairstyle suggestions		Logged-in User	
	GET		/api/recommend/outfits		Get outfit suggestions		Logged-in User	
	GET		/api/recommend/colors		Get color palettes		Logged-in User	
	GET		/api/history		Get recommendation history		Logged-in User	
	POST		/api/feedback/add		Submit feedback		Logged-in User	
	GET		/api/admin/users		View all users		Admin only	
	GET		/api/admin/analytics		View analytics		Admin only	

8. USER FLOW

	Flow		Steps	
	-----		-----	
	User Registration		Open App → Register → Fill Details → Submit → Login	
	User Login		Open App → Login → Dashboard	
	Face Analysis		Upload Image → Face Detection → Landmark Extraction → Face Shape Prediction	
	Recommendation System		Face Analysis → AI Recommendation Engine → Display Results	
	Save History		Recommendations Generated → Save to Database	
	Admin Monitoring		Admin Login → Dashboard → Monitor Recommendations	

9. MACHINE LEARNING WORKFLOW

```text

Image Upload



Face Detection



Landmark Extraction



Face Shape Classification



Feature Analysis



Recommendation Engine



Personalized Results

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## 10. PROJECT OUTCOME

This project demonstrates:

- Computer Vision
- Machine Learning
- Recommendation Systems
- Full Stack Development
- Personalized AI Applications

Suitable for:

- Final year projects
- AI portfolio projects
- Hackathons
- Startup MVPs



- ML demonstrations

# COMPLETE PROJECT STRUCTURE

Frontend (React.js)

↓

API Layer (Axios)

↓

FastAPI Backend

↓

Machine Learning Models

↓

Database (SQLite / PostgreSQL)

Frontend contains:

- Components
- Pages
- Services
- Context API
- Hooks
- Styling

Backend contains:

- Routers
- Models
- Schemas
- Services
- Middleware
- Utilities
- Authentication
- Database Configuration

Machine Learning contains:

- Face Detection
- Landmark Extraction
- Face Shape Classification
- Recommendation Engine
- Color Analysis

# AI Personal Style & Wellness Advisor

## Complete Project File Structure & Development Guide

### PROJECT ARCHITECTURE

Frontend (React)

↓

Backend API (FastAPI)

↓

Machine Learning Models

↓

Database (SQLite/MongoDB)

### COMPLETE PROJECT FOLDER STRUCTURE

frontend/  
backend/  
ml\_models/  
uploads/  
README.md  
.gitignore

Frontend includes:

- components
- pages
- services
- context
- hooks
- assets
- styles

Backend includes:

- routers
- models
- schemas
- services
- middleware
- utils
- database.py
- config.py
- main.py

ML Models include:

- face\_detection.py
- landmark\_detection.py
- face\_shape\_classifier.py
- recommendation\_engine.py
- color\_analysis.py

### FRONTEND FILES

Navbar.jsx  
Footer.jsx  
Sidebar.jsx  
ImageUploader.jsx  
FaceResultCard.jsx  
OutfitCard.jsx  
HairstyleCard.jsx  
Loader.jsx  
ProtectedRoute.jsx  
ThemeToggle.jsx

Pages:

Home.jsx  
Login.jsx  
Register.jsx  
Dashboard.jsx  
UploadPhoto.jsx  
FaceAnalysis.jsx  
OutfitRecommendations.jsx  
HairstyleRecommendations.jsx  
ColorPalette.jsx  
Profile.jsx

History.jsx  
Settings.jsx  
About.jsx  
Contact.jsx  
AdminDashboard.jsx

#### BACKEND FILES

Routers:  
auth.py  
users.py  
analysis.py  
recommendations.py  
feedback.py  
history.py  
admin.py

Services:  
auth\_service.py  
face\_service.py  
recommendation\_service.py  
color\_service.py  
history\_service.py

Utils:  
jwt\_handler.py  
image\_utils.py  
file\_handler.py  
response\_handler.py

#### HOW FRONTEND CONNECTS TO BACKEND

Frontend uploads image  
↓  
FastAPI receives image  
↓  
ML model processes image  
↓  
Recommendation engine generates results  
↓  
Backend returns JSON response  
↓  
Frontend displays results

#### AUTHENTICATION FLOW

User Registers  
↓  
Password Hashed  
↓  
Saved In Database  
↓  
User Logs In  
↓  
JWT Token Generated  
↓  
Frontend Stores Token

#### HOW TO RUN FRONTEND

```
cd frontend
npm install
npm run dev
```

Frontend:  
<http://localhost:5173>

#### HOW TO RUN BACKEND

```
cd backend
uvicorn app.main:app --reload
```

Backend:  
<http://localhost:8000>

Swagger Docs:  
<http://localhost:8000/docs>

#### HOW TO CONNECT REACT WITH FASTAPI

Install axios:

```
npm install axios
```

#### DEVELOPMENT ROADMAP

PHASE 1 - Setup project folders  
PHASE 2 - Build frontend pages  
PHASE 3 - Build FastAPI backend  
PHASE 4 - Build face detection  
PHASE 5 - Build face shape prediction  
PHASE 6 - Build recommendation engine  
PHASE 7 - Connect frontend + backend + ML  
PHASE 8 - Add authentication  
PHASE 9 - Add database  
PHASE 10 - Deploy project online

#### FINAL RESULT

Full Stack Application  
Computer Vision System  
AI Recommendation System  
Personalized Fashion Assistant  
Portfolio-Level ML Project  
Startup MVP Prototype